

10.4 Joseph Weber, demonstrating the piezoelectric crystals glued around the middle of his aluminum bar; ca. 1973. Gravitational waves should drive the bar's end-to-end vibrations, and those vibrations should squeeze the crystals in and out so they produce oscillating voltages that are detected electronically. [Photo by James P. Blair, courtesy the National Geographic Society.]

Weber's sensor was based on the *piezoelectric effect*, in which certain kinds of materials (certain crystals and ceramics), when squeezed slightly, develop electric voltages from one end to the other. Weber would have liked to make his bar from such a material, but these materials were far too expensive, so he did the next best thing: He made his bar from aluminum, and he then glued piezoelectric crystals around the bar's middle (Figure 10.4). As the bar vibrated, its surface squeezed and stretched the crystals, each crystal developed an oscillating voltage, and Weber strung the crystals together one after another in an electric circuit so their tiny oscillating voltages would add up to a large enough voltage for electronic detection,